

Java vs. JavaScript: Core Concept Cheat Sheet

A quick reference guide for variable syntax, typing systems, concatenation mechanics, and shared naming identifiers.

1. Variable Naming Rules (Identical in Both)

Both Java and JavaScript share the exact same rules for naming variables:

- **Allowed Characters:** Names can contain letters, digits, underscores (`_`), and dollar signs (`$`).
- **First Character:** Names must begin with a letter, `$`, or `_`.
- **Formatting & Whitespace:** Names should start with a lowercase letter (camelCase format), and **cannot** contain spaces.
- **Case-Sensitivity:** Names are strictly case-sensitive. `myVar` and `myvar` are completely different variables.
- **Reserved Words:** Language keywords (like `int`, `boolean`, `let`, or `class`) cannot be used as names.

2. Variable Declarations & Data Types

Feature	Java (Static Typing)	JavaScript (Dynamic Typing)
Declaration Syntax	Requires explicit type: <code>int x = 5;</code>	Uses keywords: <code>let x = 5;</code> or <code>const y = 2;</code>
The Single Quote Rule	Strictly a char type. Enclosed in <code>' '</code> (e.g., <code>'a'</code>). Evaluates as its numeric decimal value (ASCII 97).	No character type. Single quotes <code>' '</code> and double quotes <code>" "</code> both create standard text strings.
Primitive Types	<code>int</code> , <code>char</code> , <code>float</code> , <code>boolean</code> , <code>double</code> , etc.	<code>number</code> , <code>string</code> , <code>boolean</code> , <code>undefined</code> , <code>null</code> , etc.

3. The Concatenation Domino Effect (The Golden Rule)

When chained using the `+` operator, expressions are evaluated from left to right like a book. **If a string is involved first, the entire remaining statement collapses into a single string.**

Without Parentheses (Left-to-Right Domino)

```
"The sum is " + 5 + 2; // Result: "The sum is 52"
```

1. `"The sum is " + 5` turns into the string `"The sum is 5"`.
2. `"The sum is 5" + 2` turns into the final string `"The sum is 52"`.

With Parentheses (Math First Exception)

```
"The sum is " + (5 + 2); // Result: "The sum is 7"
```

1. Parentheses `()` hold the highest priority; `5 + 2` executes first as math to yield `7`.
2. The text meets the number `7` via `+`, converting the final output to the string `"The sum is 7"`.